CMP-310 Operating Systems | Homework 1 | Spring 2025

Due Date: February 21 at 11:50 PM **(10 points)**

Question 1: The Collatz conjecture concerns what happens when we take any positive integer n and apply the following algorithm:

$$n = \begin{cases} n/2, & \text{if n is even} \\ 3 \times n + 1, & \text{if n is odd} \end{cases}$$

The conjecture states that when this algorithm is continually applied, all positive integers will eventually reach 1. For example, if n = 35, the sequence is

35, 106, 53, 160, 80, 40, 20, 10, 5, 16, 8, 4, 2, 1

Write a C program using the fork() system call that generates this sequence in the child process. The starting number will be provided from the command line. For example, if 8 is passed as a parameter on the command line, the child process will output 8, 4, 2, 1. Because the parent and child processes have their own copies of the data, it will be necessary for the child to output the sequence. Have the parent invoke the wait() call to wait for the child process to complete before exiting the program. Perform necessary error checking to ensure that a positive integer is passed on the command line.

Question 2: Write a C program that creates a child process to count the number of vowels in a given string. The program should perform the following tasks:

- Input Handling: The program should accept a string as a command-line argument. If no string is provided or if the string is empty, it should print an error message and exit.
- Create a Child Process: Use fork() to create a child process.
- Execute the Vowel Count Program: In the child process, use execlp() to run a separate program called vowel_count that counts the number of vowels in the provided string.
- Vowel Count Program (vowel_count.c): This program should read the string passed to it as a command-line argument, count the vowels (a, e, i, o, u, both uppercase and lowercase), and print the total count.
- Parent Process: The parent process should wait for the child process to complete using wait() before exiting.

Example:

To compile:

gcc main.c -o main

gcc vowel_count.c -o vowel_count

To run:

./main "Hello World"

Number of vowels: 3

Guidelines:

Main Program:

- Ensure to check for proper command-line input.
- Create the child process and handle any errors from fork().
- Pass the string to the child process using execlp().

Vowel Count Program:

- Implement a loop to count vowels.
- Print the result in the format specified.

Hints:

- In main.c, use fork() to create a child process.
- In vowel_count.c, read the string and loop through it to count vowels.
- Use wait() in the main program to wait for the child to finish.

Submission Requirements:

- Submit the following files:
  - collatz.c (for Question 1)
  - main.c (for Question 2)
  - vowel_count.c (for Question 2)
- Execution Instructions File:
  - Include a text file named README.txt with the following:
    - Compilation commands for each program.
    - Instructions on how to run each program, including sample inputs and expected outputs.
- Screenshot Evidence:
  - Include screenshots showing:
    - Successful compilation of each program.
    - Output of the programs when run with sample inputs.
- Archive Your Submission:
  - Zip all files (main.c, vowel_count.c, collatz.c, README.txt, and screenshots) into a single file named: CMP310_HW1_YourName.zip
- Upload Instructions:
  - Submit your zipped file on iLearn under the Homework 1 submission link before February 21 at 11:50 PM.

Good luck!